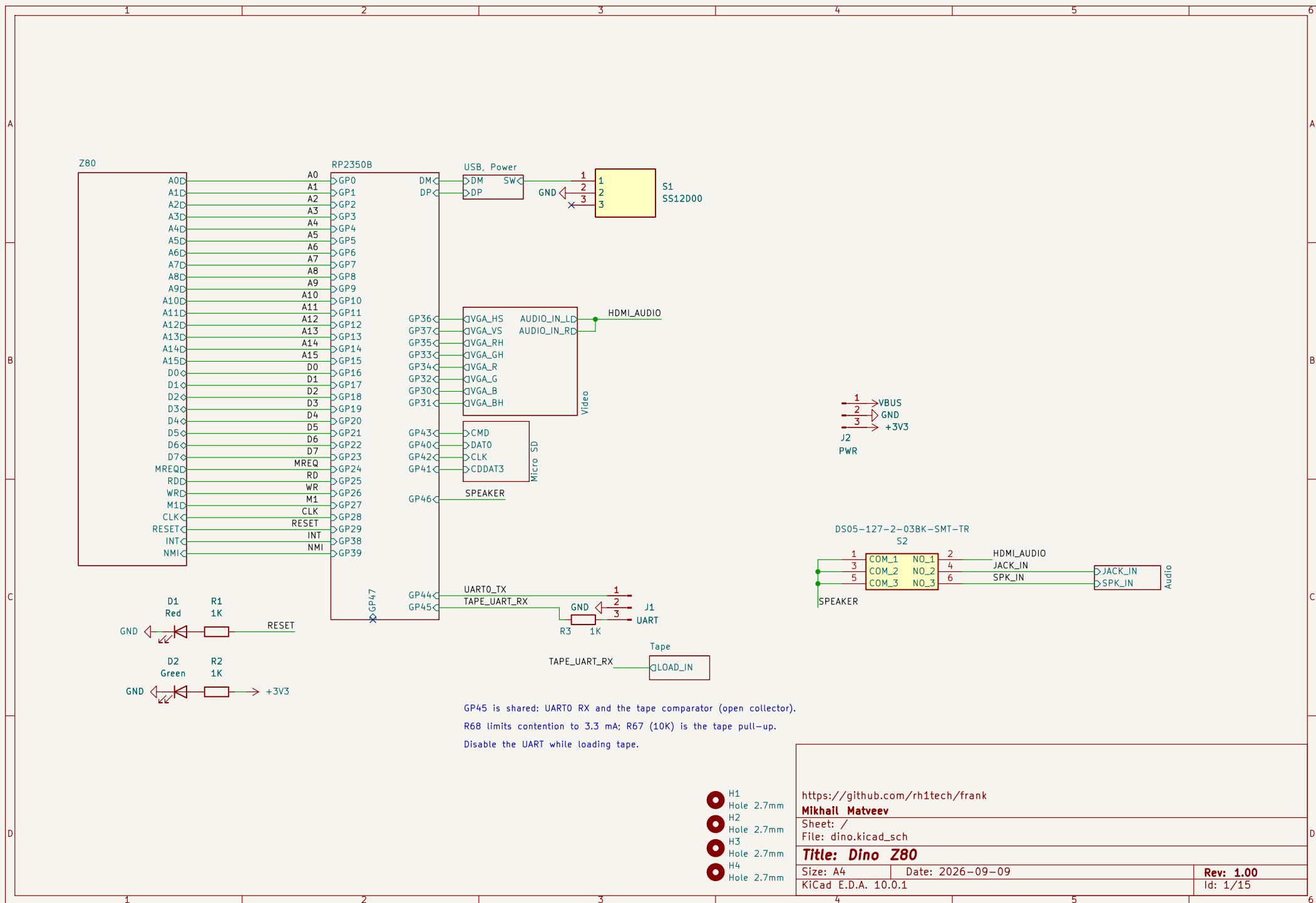




The RP2350B generates CLK and can stop it, so the CPU is frozen whenever the emulator needs time. That removes all wait-state logic: /WAIT is tied high. U2 must be a CMOS, FULLY STATIC part – an NMOS Z8400 loses state when the clock stops.

CLK goes through U17 (74HCT125): HCT inputs are TTL (V_{IH} 2.0 V) so GP28's 3.3 V drives it, and its push-pull output swings to 5 V, meeting the Z80's V_{IHC} = V_{cc} - 0.6 = 4.4 V. Every parameter is now inside spec.

R4 holds the buffer input low while GP28 is hi-Z, so the CPU clock parks low. /NMI is driven, not tied: stopping the clock freezes the CPU but cannot read its registers, so a snapshot save has to vector in through NMI (the Z80 will refuse INT after DI).

V_{cc} is VBUS (5 V), so the Z80 runs fully in Zilog spec and at its rated speed. VBUS is downstream of the Q3 load switch and feeds the 3V3 buck, so the CPU and the RP2350B power up and down together – no sequencing hazard.

The bus connects straight to GPIO with no level shifters: RP2350B GP0–GP39 are 5 V fault-tolerant, and the Z80's data and control inputs are TTL (V_{IH} 2.2 V), which a 3.3 V GPIO drives cleanly.

/IORQ, /RFSH, /HALT and /BUSACK are deliberately unconnected.

/IORQ is redundant: /MREQ low = memory; /MREQ high with /RD or /WR low = I/O;

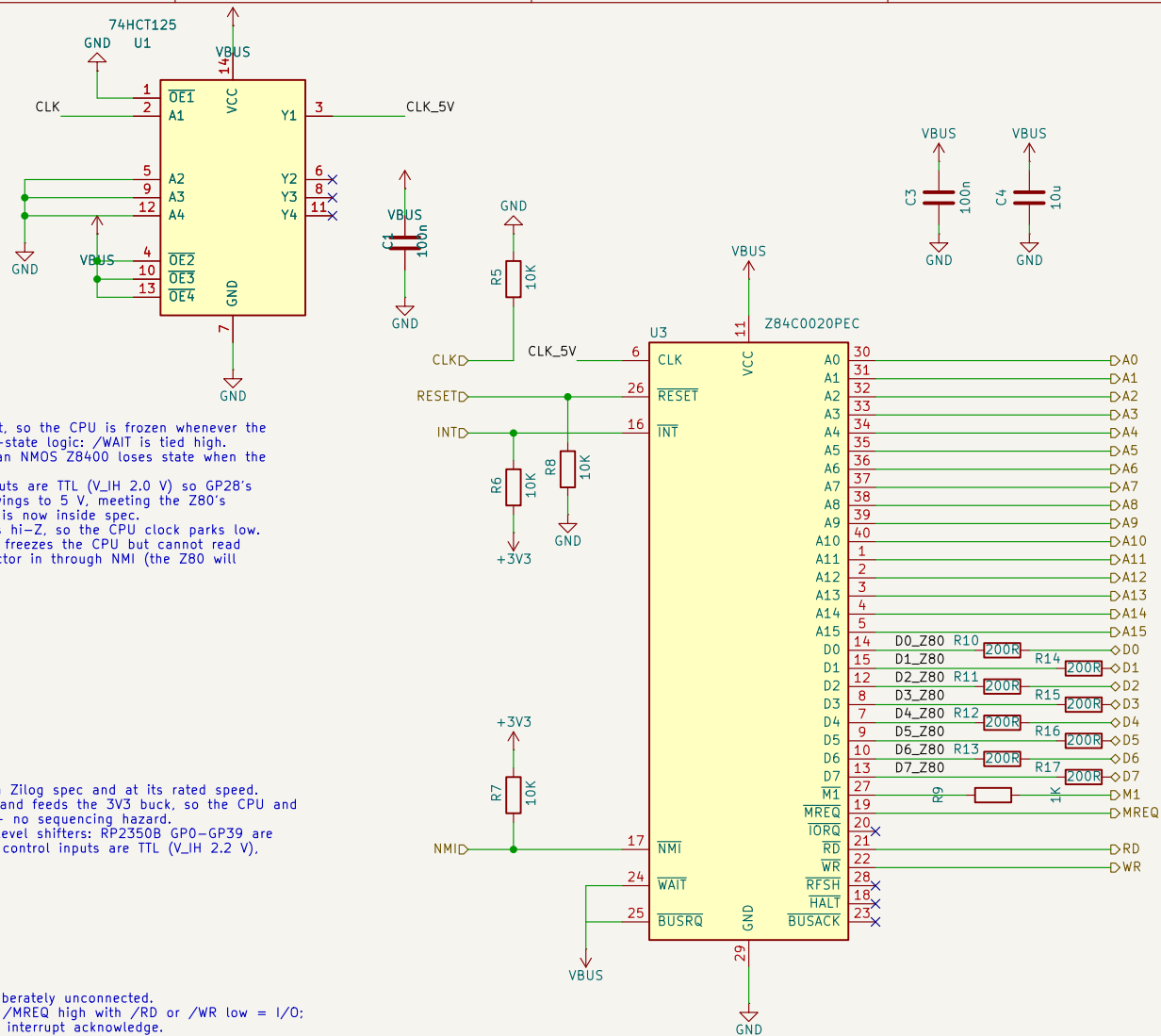
/M1 low with /MREQ, /RD and /WR high = interrupt acknowledge.

/RFSH is redundant: refresh cycles leave both /RD and /WR inactive.

R7 holds the CPU in reset while GP29 is hi-Z during RP2350B boot.

R8–R15 (200R) limit fault current if firmware ever drives the data bus against the Z80 – that contention is now 5 V vs 3.3 V, so keep them fitted.

R69 (1K) limits /M1 during the ~1 ms at power-on when VBUS is up but the 3V3 buck is still soft-starting: /M1 is a driven output and is high in reset.



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Mikhail Matveev

Sheet: /Z80/

File: z80.kicad_sch

Title: Dino Z80

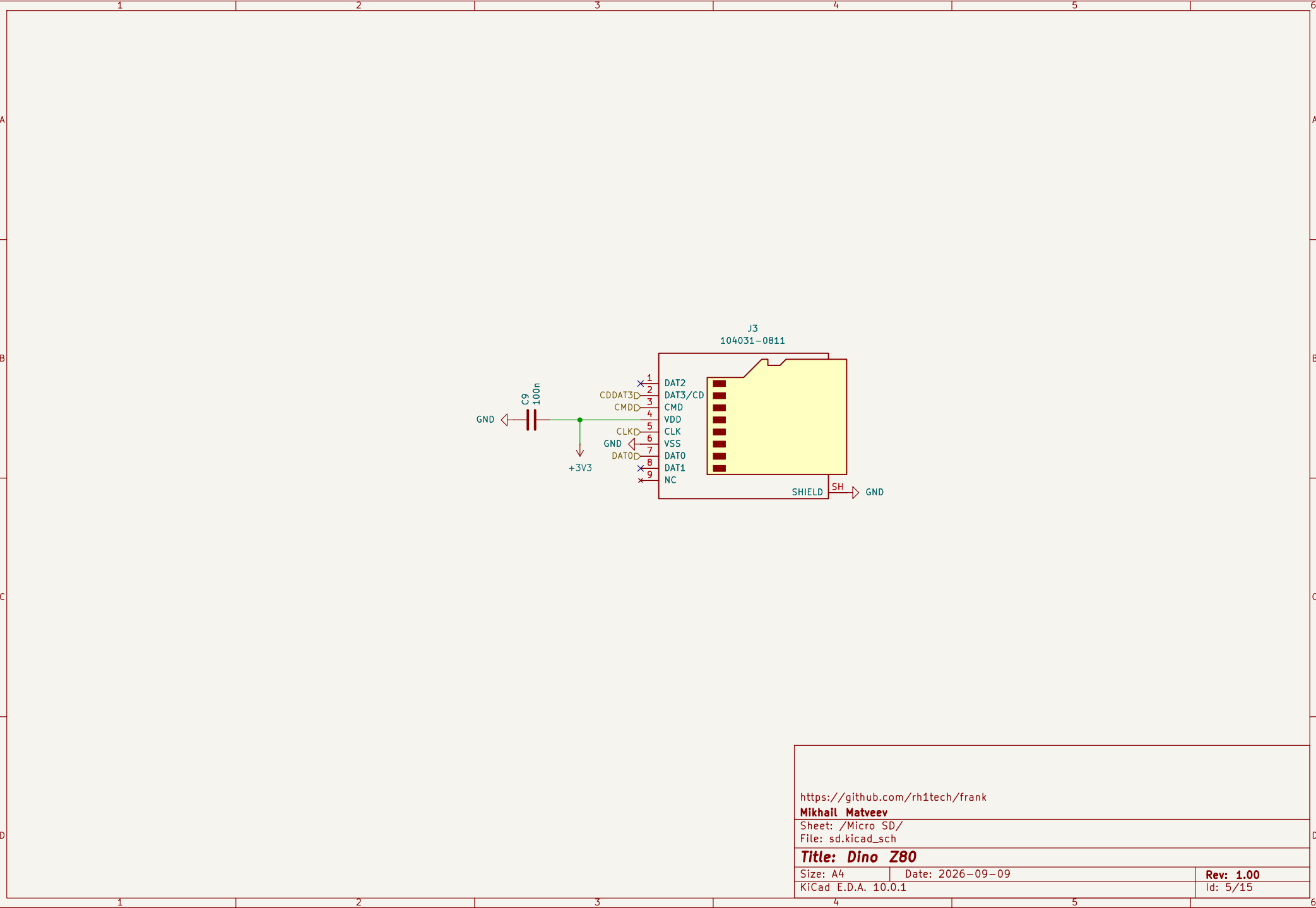
Size: A4

Date: 2026-09-09

Rev: 1.00

KiCad E.D.A. 10.0.1

Id: 2/15



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Sheet: /Micro SD/

File: sd.kicad_sch

Title: Dino Z80

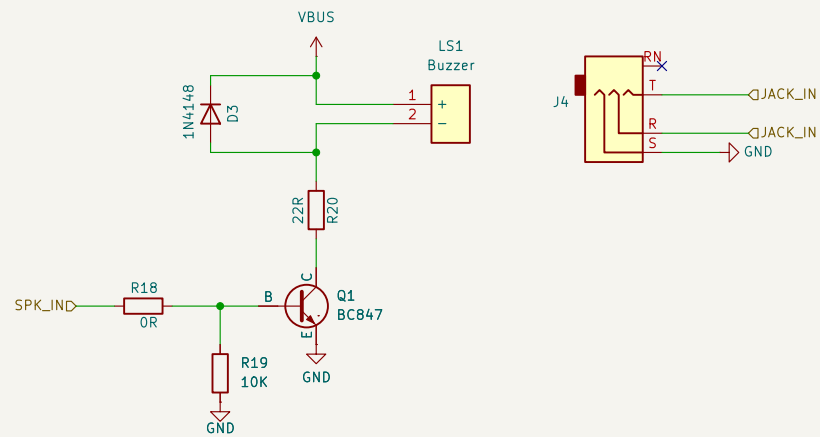
Size: A4

Date: 2026-09-09

Rev: 1.00

KiCad E.D.A. 10.0.1

Id: 5/15



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Sheet: /Audio/

File: pc_speaker.kicad_sch

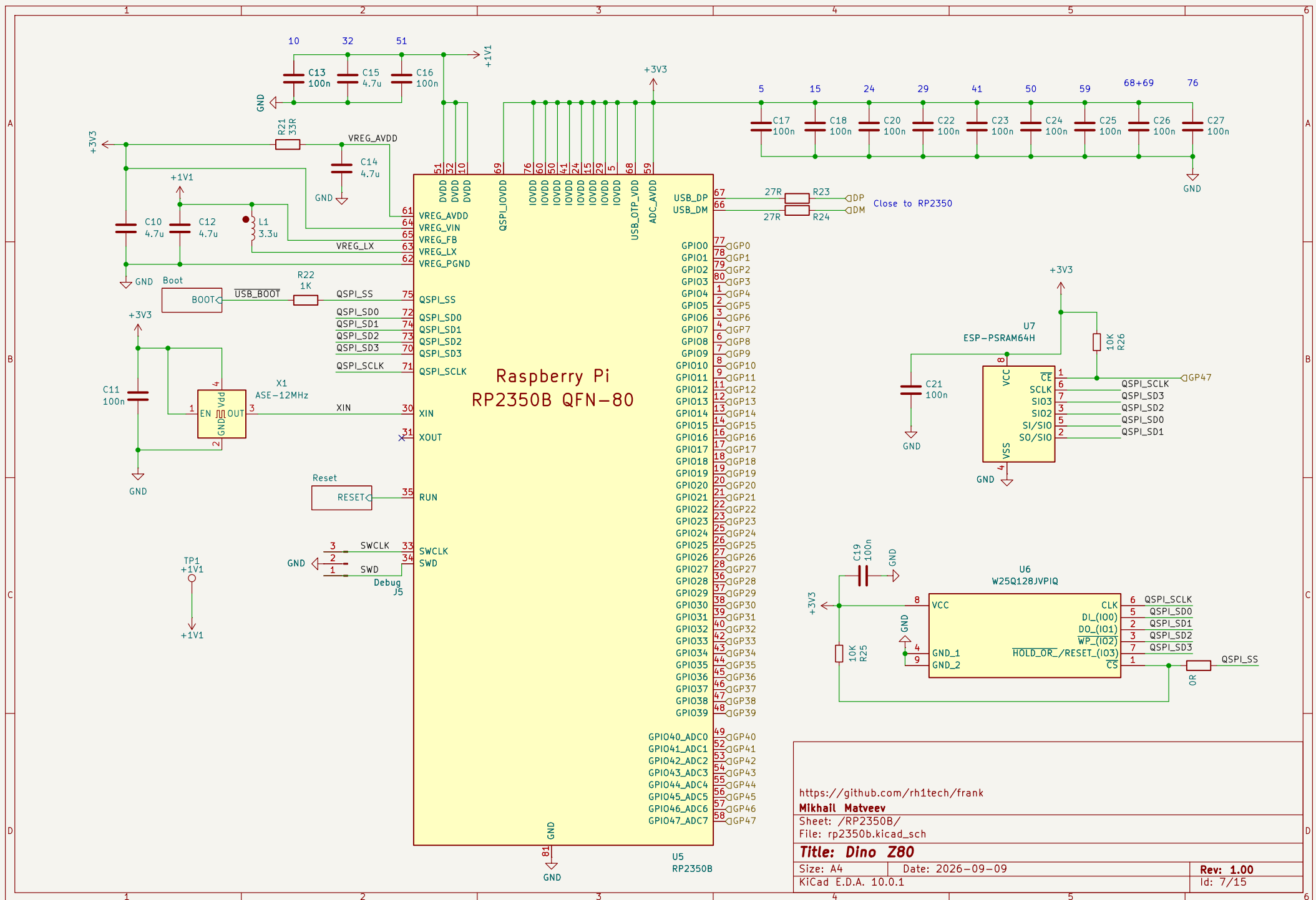
Title: Dino Z80

Size: A4 Date: 2026-09-09

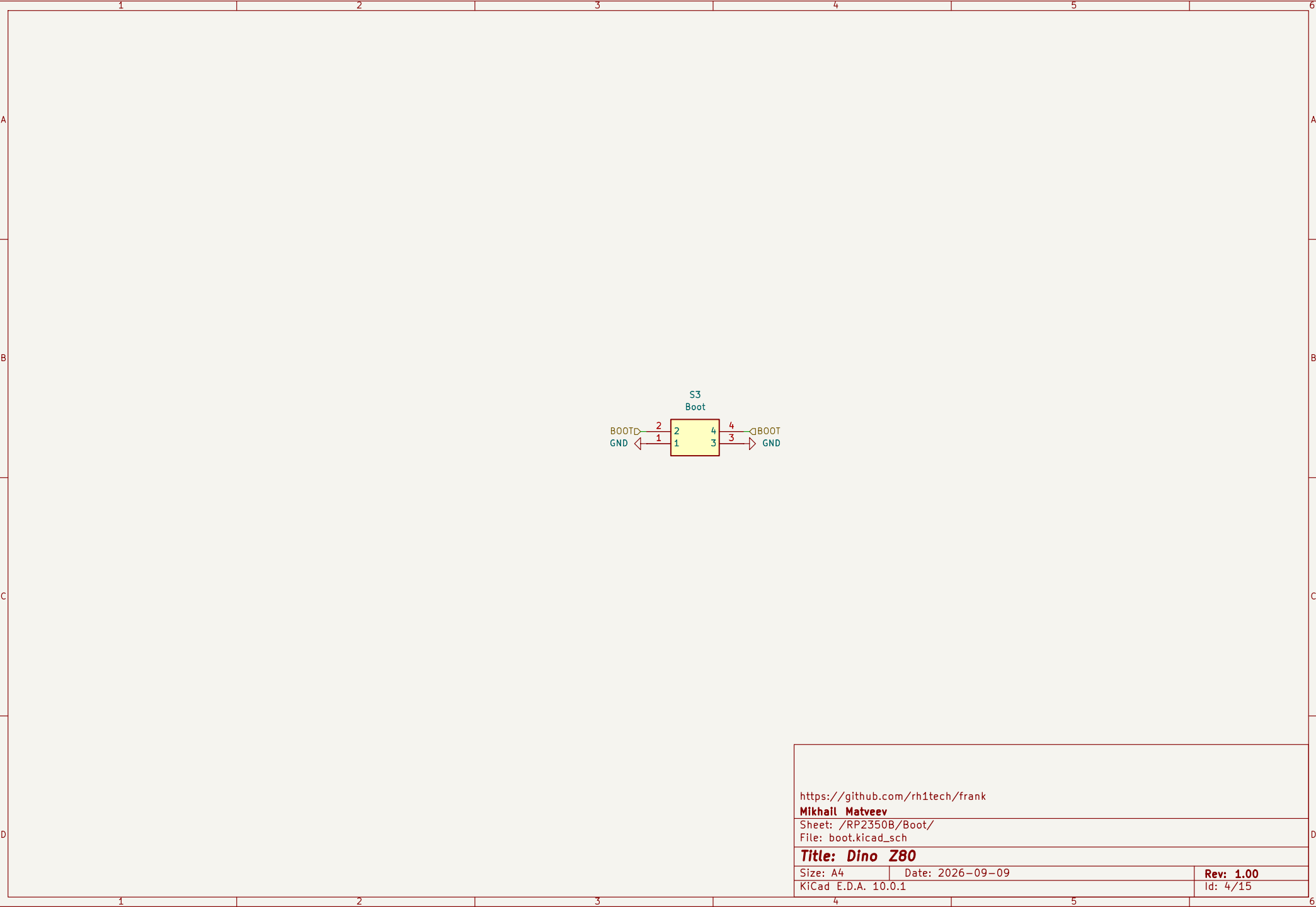
KiCad E.D.A. 10.0.1

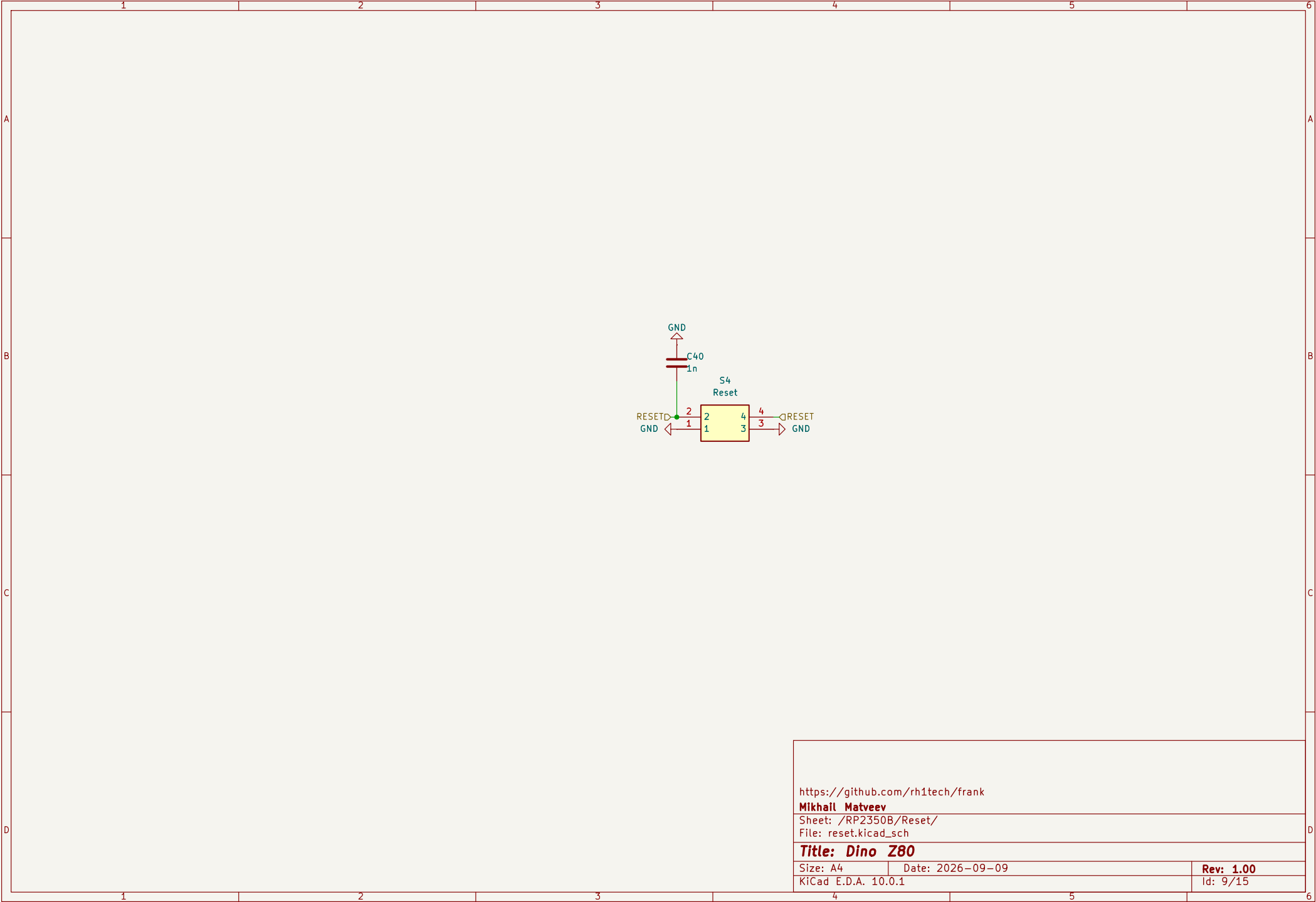
Rev: 1.00

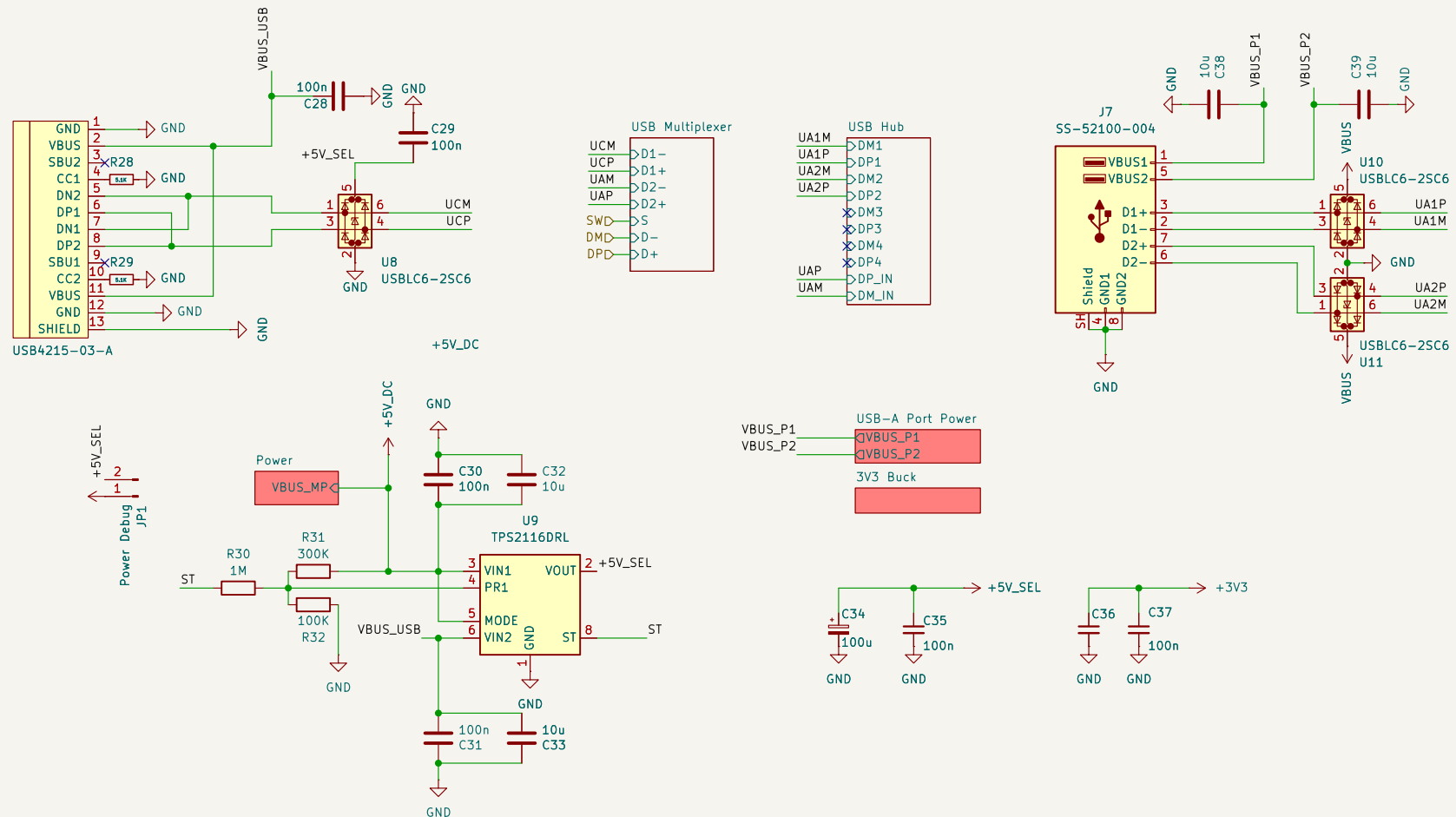
Id: 5/15

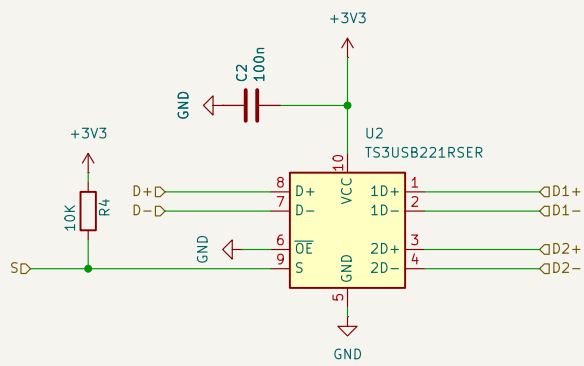


Id: 7/15

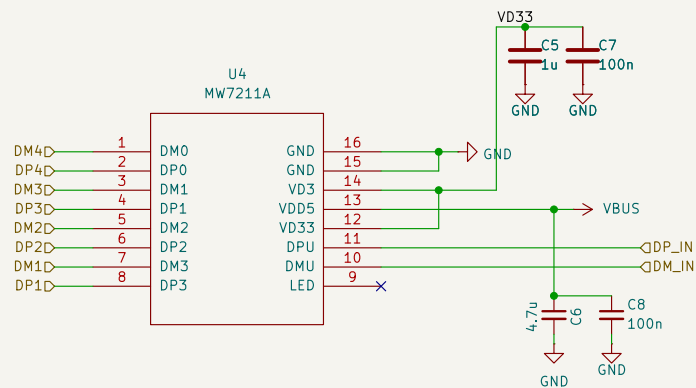








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Sheet: /USB, Power/USB Multiplexer/ File: usb_muxiplexer.kicad_sch		
Title: Dino Z80		
Size: A4	Date: 2026-09-09	Rev: 1.00
KiCad E.D.A. 10.0.1	Id: 2/15	



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Sheet: /USB, Power/USB Hub/

File: usb_hub.kicad_sch

Title: Dino Z80

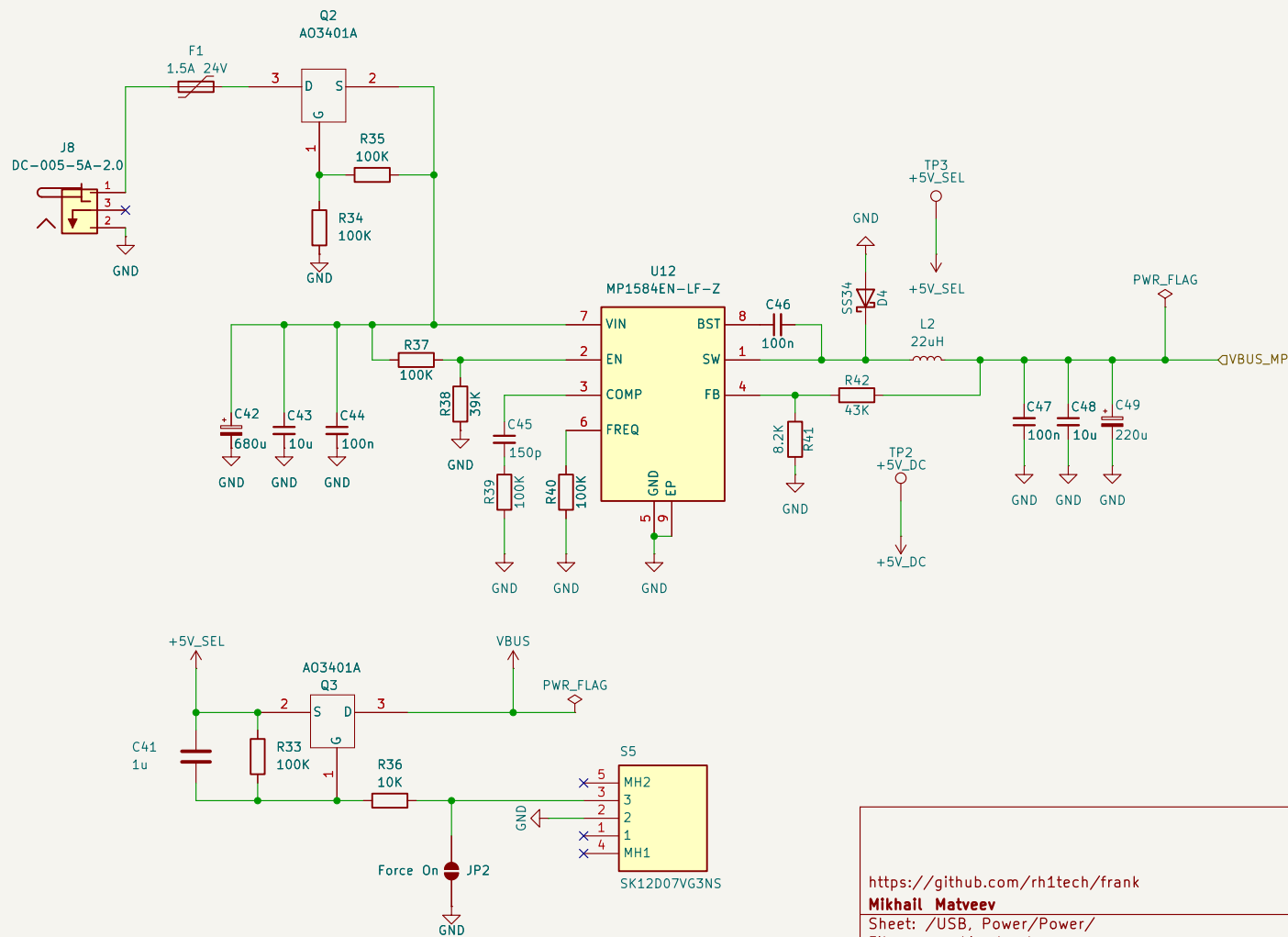
Size: A4

Date: 2026-09-09

Rev: 1.00

KiCad E.D.A. 10.0.1

Id: 3/15



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Mikhail Matveev

Sheet: /USB, Power/Power/

File: power.kicad_sch

Title: Dino Z80

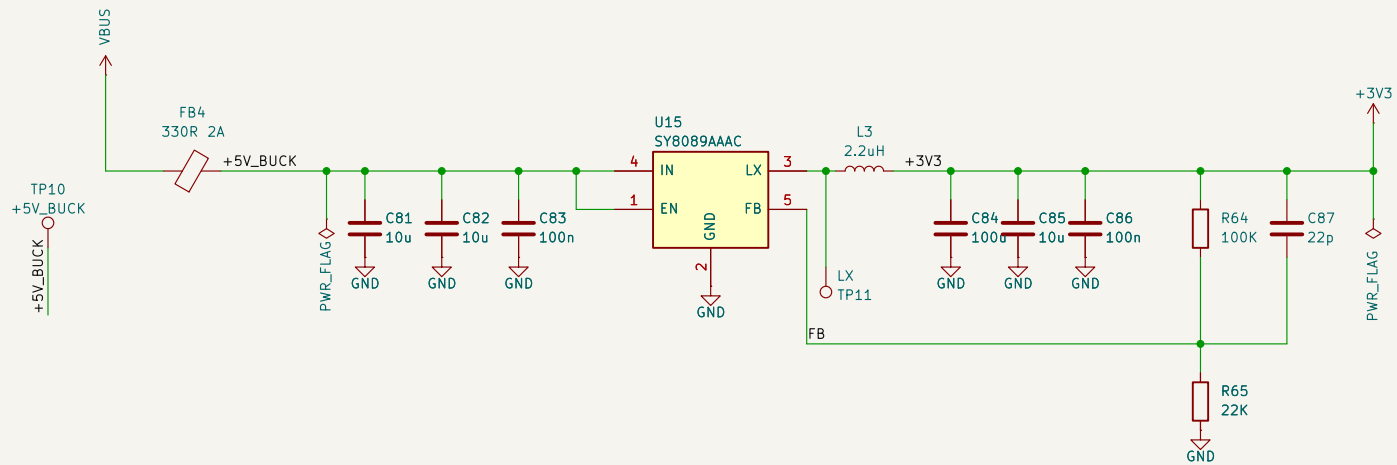
Size: A4

Date: 2026-09-09

Rev: 1.00

KiCad E.D.A. 10.0.1

Id: 10/15



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Sheet: /USB, Power/3V3 Buck/

File: buck3v3.kicad_sch

Title: Dino Z80

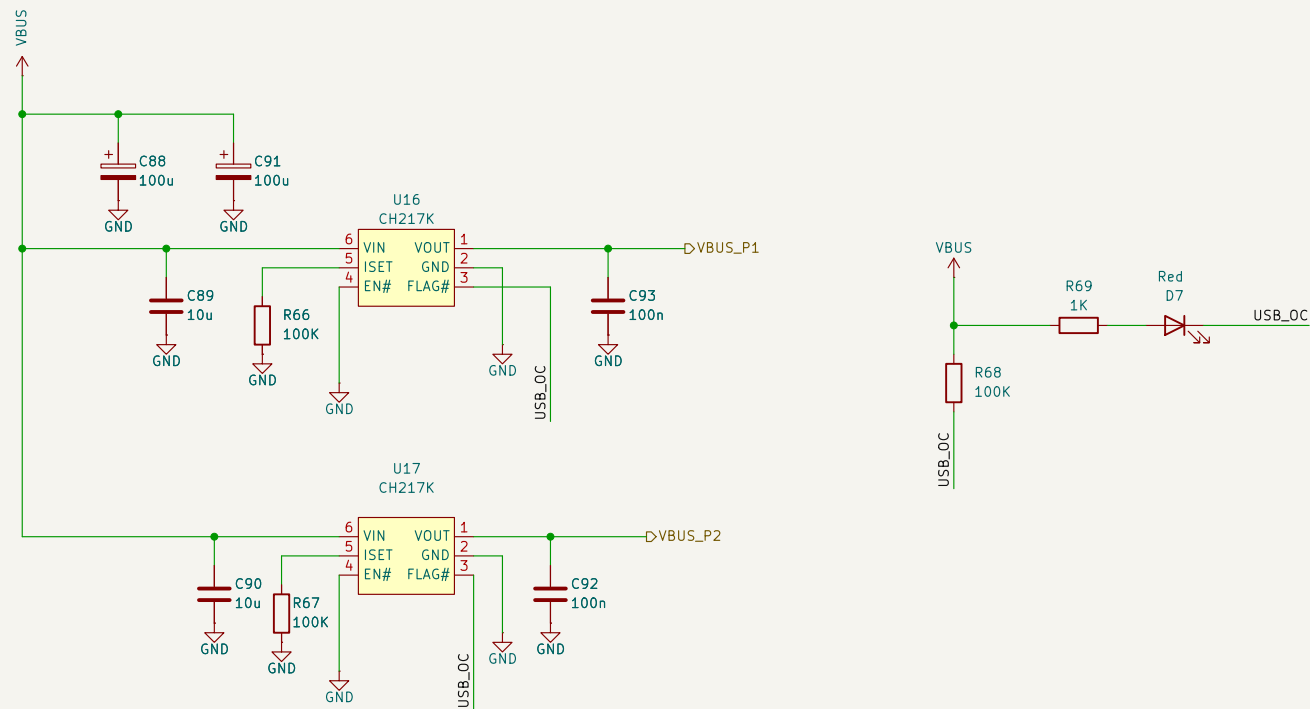
Size: A4

Date: 2026-09-09

Rev: 1.00

KiCad E.D.A. 10.0.1

Id: 17/15



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Mikhail Matveev

Sheet: /USB, Power/USB-A Port Power/

File: usb_a_power.kicad_sch

Title: Dino Z80

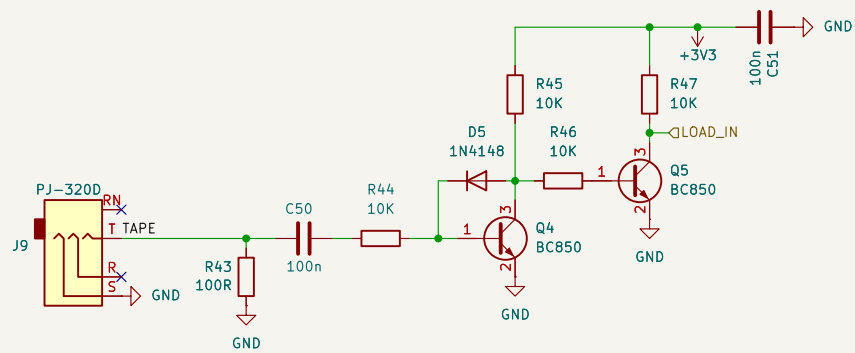
Size: A4

Date: 2026-09-09

Rev: 1.00

KiCad E.D.A. 10.0.1

Id: 18/15



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Mikhail Matveev

Sheet: /Tape/

File: tape.kicad_sch

Title: Dino

Size: A4

Date: 2026-09-09

Rev: 1.00

KiCad E.D.A. 10.0.1

Id: 15/15

